INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

CHM201: SPECTROSCOPIC AND OTHERR PHYSICAL METHODS

DURATION: 1 HR

MARKS: 15 **MID-SEMESTER – I (2025-26)**

Constants: $N = 6.023*10^{23}$ mol^{-1}; $k = 1.381*10^{-23}$ JK^{-1}; $h = 6.626*10^{-34}$ Js; $c = 3*10^{8}$ ms^{-1} ;
Rel. atomic weight of H = 1.000 ; absolute mass of H-atom = $1.67343*10^{-27}$ kg.

1. The B of a symmetric rotor 20 cm^{-1}. (a) Please estimate the most populated J^{th} state for the rotor at T = 300 K. (b) Qualitatively draw all possible directions of the angular momentum corresponding to the J_{max} and their corresponding magnitudes on the Laboratory-axis. **2+ 2 + 1 = 5**

2. Which of the following molecules may show a pure rotational microwave spectrum: (i) H_2, (ii) HCl, (iii) CH_4, (iv) H_2O. **2**

3 Draw schematically the pure rotational energy levels of rigid $^{12}C^{16}O$ and compare it with $^{13}C^{16}O$. Extend this comparison for non-rigid $^{13}C^{16}O$ on the same energy diagram. **3**

4 Mark of 1 each question. You get +1 for correct answer(s) or -1 for the wrong one. Write the question number and the full answer given in the options. **5x1**

a) In FT-IR spectroscopy, we do Fourier transform of the data as
(i) Temporal function to frequency function
(ii) Interferrogram to frequency function
(iii) Frequency function to Interferrogram
(iv) Temporal function to wavelength function

b) The lifetime of a state that gives rise to a line of width 1 cm^{-1} is
i) 5 ps
ii) 5 ns
iii) 10 ns
iv) 1 ns

c) The peak maxima for any rotational spectrum
i) appears at J =0 to J = 1 transition always
ii) Is dependent on molecules under probe
iii) Is dependent on the ground state of v
iv) Is random

d) Blue sky is fact of
i) Mie scattering
ii) Raman scattering
iii) Rayleigh scattering
iv) Tyndall effect

e) Absorbance (A) higher than 2 is
i) Desirable as it makes the measurement more efficient
ii) Not desirable as only a small fraction of light transmitted to detector
iii) Desirable otherwise detector cannot detect molecules
iv) Not desirable as detector goes blind with such a high intensity of light